

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)			Total percentage of ^{29}Si and ^{30}Si is $100 - 92.2 = 7.8$ $\% \text{ of } ^{29}\text{Si} = \frac{2 \times 7.8}{3} = 5.2$ and $\% \text{ of } ^{30}\text{Si} = \frac{1 \times 7.8}{3} = 2.6$ (1) $\therefore A_r = \frac{(28 \times 92.2) + (29 \times 5.2) + (30 \times 2.6)}{100}$ (1) $\therefore A_r = \frac{2582 + 151 + 78}{100} = \frac{2811}{100} = 28.1$ (1) Answer only – no mark ecf possible			1		1	
	(b)	(i)		Tetrahedral	1			1		
		(ii)		There are no free electrons or ions to carry the charge	1			1		
		(iii)		Any of the following - There are no electronegativity differences in the Si—Si bond - All the bonding electrons are shared equally between the four Si atoms - Si cannot lose or gain 4 electrons			1	1		